Pre-Fertilisation: Structures and Events

- Several hormonal and structural changes result in the development of a flower.
- Inflorescences bear the flower buds, and then the flowers.
- Flowers are the reproductive parts of a plant.
- In the flowers, the androecium (male reproductive part) and the gynoecium (female reproductive part) develop.

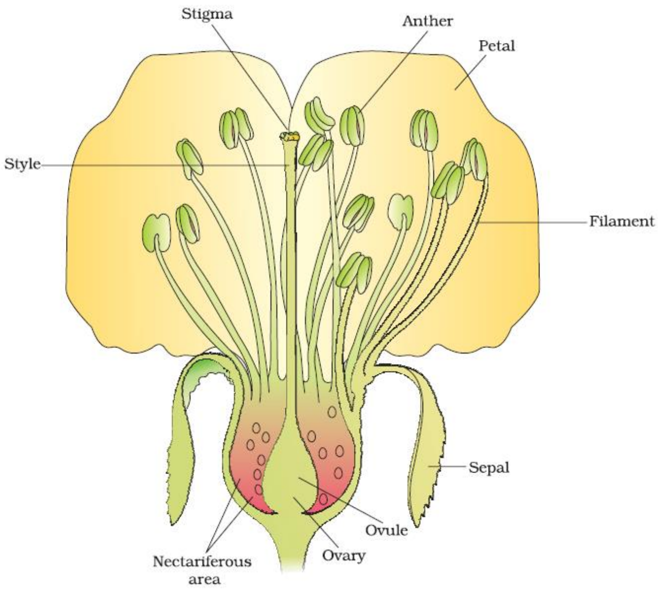


Figure: Diagrammatic representation of L.S. of a flower

Androecium

- The androecium consists of whorls of stamen.
- The stamen consists of the **filament** (long and slender stalk) and **anther** (bilobed structure).
- Proximal end of filament is attached to the thalamus or to the petal.

Anther:

- A typical anther is bilobed and each lobe is dithecous (consists of two theca).
- Theca are separated by a longitudinal groove running lengthwise.
- The microsporangia are located at the corners, two in each theca. They further develop to form pollen sacs, which contain the pollen grains.

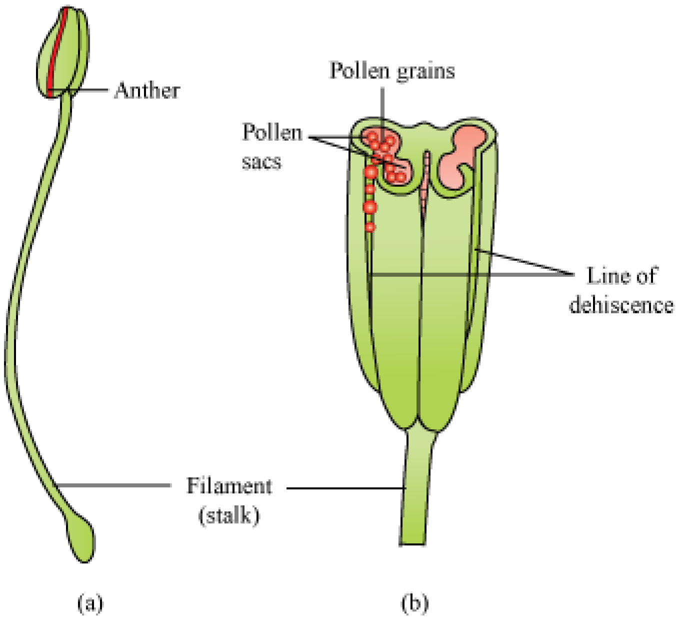


***Figure:** (a) A typical stamen; (b) three-dimensional cut section of an anther*

Structure of microsporangium

- The microsporangium is surrounded by four wall layers (epidermis, endothecium, middle layers, and tapetum).
- The outer three layers are protective and help in dehiscence of anther to release the pollen grains. The tapetum provides nourishment to the developing pollen grains.
- In the young anther, the sporogenous tissue forms the centre of each microsporangium.

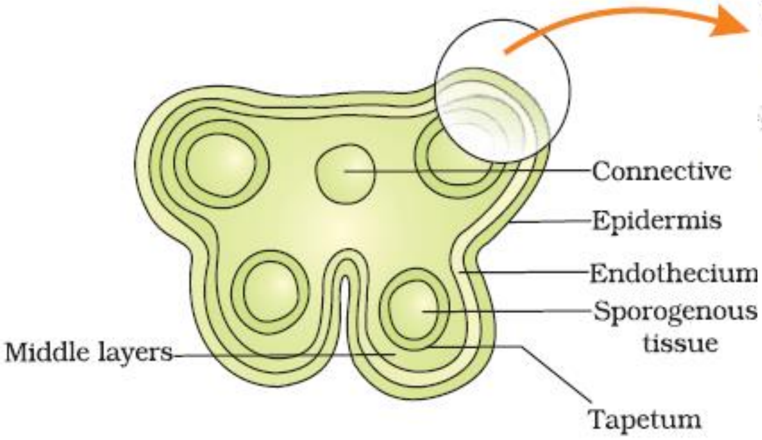


Figure (a) Transverse section of a mature anther

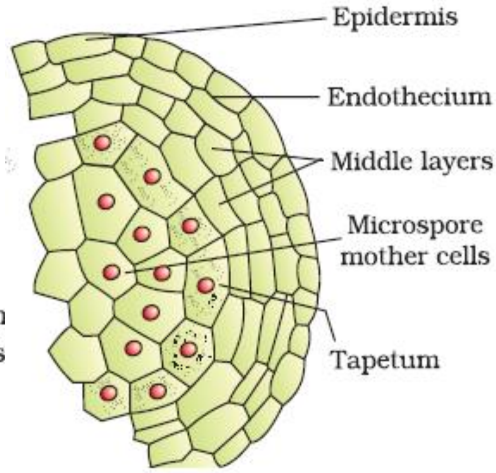


Figure (b) Enlarged view of one microsporangium

Microsporogenesis

- It is the process of formation of microspore from PMC (Pollen Mother Cells).
- As development occurs in the anther, the sporogenous tissue undergoes meiosis to form microspore tetrad.
- Each cell of sporogenous tissue has capacity to give rise to a tetrad. Hence, each cell is a potential pollen or PMC.
- As the anther matures, the microspores get detached from each other and develop into pollen grains or microspore.
- Mature microspores or pollen grains are released with the dehiscence of anther.

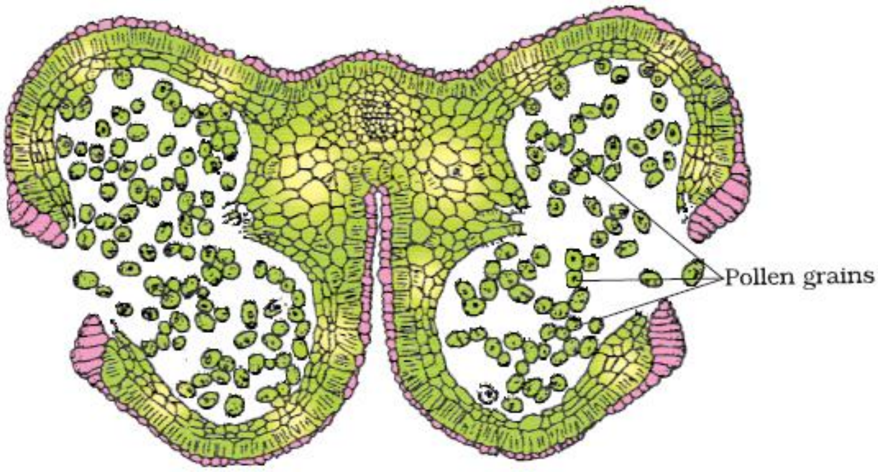
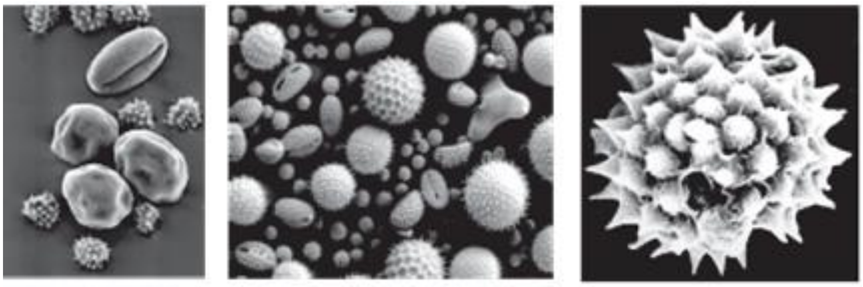


Figure: A dehisced anther

Figure: Enlarged view of a pollen grain tetrad

Pollen grains

- Represent the male gametophyte and are spherical (20-25 micrometer in diameter), having a two-layered wall:
  - Exine (outer) – Hard layer made of sporopollenin, which is extremely resistant and can withstand high temperatures, acidic and alkaline conditions, and enzymes. It has prominent appertures called germ pores where sporopollenin is absent
  - Intine (inner) – Thin and continuous layer made up of cellulose and pectin



Scanning electron micrpgraphs of a few pollen grains

NOTE: - Pollen grains are well preserved as fossils because of the presence of sporopollenin.

- Mature pollen grain contains two cells:

  - Vegetative cell – Large with irregular nucleus, contains food reserves
  - Generative cell – Small, spindle shaped and floats in the cytoplasm of the vegetative cell

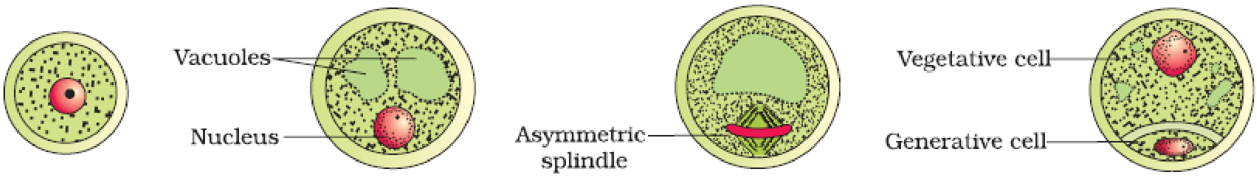


Figure: *Stages of a microspore maturing into a pollen grain*

- In 60% of the angiosperms, pollen grains are shed at 2-celled stage while in others generative cell undergoes mitosis to form two male gametes (3-celled stage) before pollen grains are shed.
- The viability of pollen grains after they are shed depends upon temperature and humidity.
- It ranges from 30 minutes to few months.

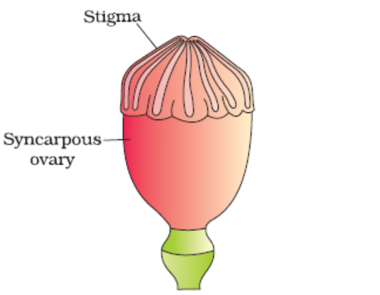
NOTE –

- **Viability -** In some cereals such as rice and wheat, pollen grains lose viability within 30 minutes of their release. In some members of Rosaceae, Leguminaceae and Solanaceae, they maintain viability for months.
- **Storage -** It is possible to store pollen grains of a large number of species for years in liquid nitrogen (-196^{0}C). Such stored pollen can be used as pollen banks, similar to seed banks, in crop breeding programmes.
- **Allergy -** Pollen grains of many species cause severe allergies and bronchial afflictions in some people often leading to chronic repiratory disorders – asthma, bronchitis, etc. It may be mentioned that *Parthenium* or carrot grass that came into India as a contaminant with imported wheat has become ubiquitous in occurrence and causes pollen allergy.
- **Food supplements -** Pollen grains are rich in nutrients and used as food supplements.

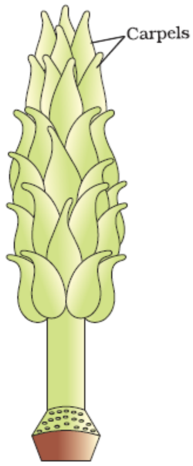
In western countries, a large number of pollen products in the form of tablets and syrups are available in the market. Pollen consumption has been claimed to increase the performance of athletes and race horses.

Gynoecium

- The gynoecium represents the female reproductive part of a flower.
- It may be mono-carpellary (one pistil) or multi-carpellary (many pistils). In multi-carpellary, the pistils may be fused in one (syncarpous) example – Papaver or free (apocarpous) example – Michelia.



Multicarpellary, syncarpous pistil of papaver



A multicarpellary, apocarpous gynoecium of Michelia

- Each pistil consists of:
  - **Stigma** – Receives the pollen grains
  - **Style** – Elongated, slender part below the stigma
  - **Ovary** – Bulged basal part containing the placenta, which is located inside the ovarian cavity (locule)
  - The placenta is located in locule and contains the megasporangia or ovules.

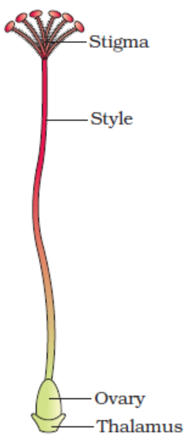


Figure: *A dissected flower of Hibiscus showing pistil (other floral parts have been removed)*

The Megasporangium (Ovule)

- The ovule is attached (one in wheat, paddy, mango and many in papaya, water melon, orchids) to the placenta by the stalk callled **funicle**. The junction of the ovule and the funicle is called **hilum**.
- Each ovule has one or two protective layers, called **integuments**, which cover the rest of the ovule, except for a small opening called **micropyle**.
- The **chalaza** lying on the opposite side of the micropyle end represents the basal part of the ovule.
- **Nucellus** is present within the integuments and contains reserved food. The **embryo sac** or female gametophyte is located within the nucellus.

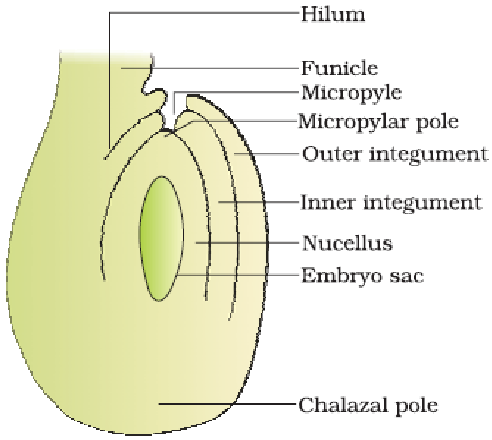


Figure: *A diagrammatic view of a typical anatropous ovule*

Megasporogenesis

- The **megaspore mother cell** (MMC) gets converted into megaspores by the process of megasporogenesis.
- The MMC is large and contains a dense cytoplasm and a prominent nucleus. It undergoes meiosis to produce four megaspores.

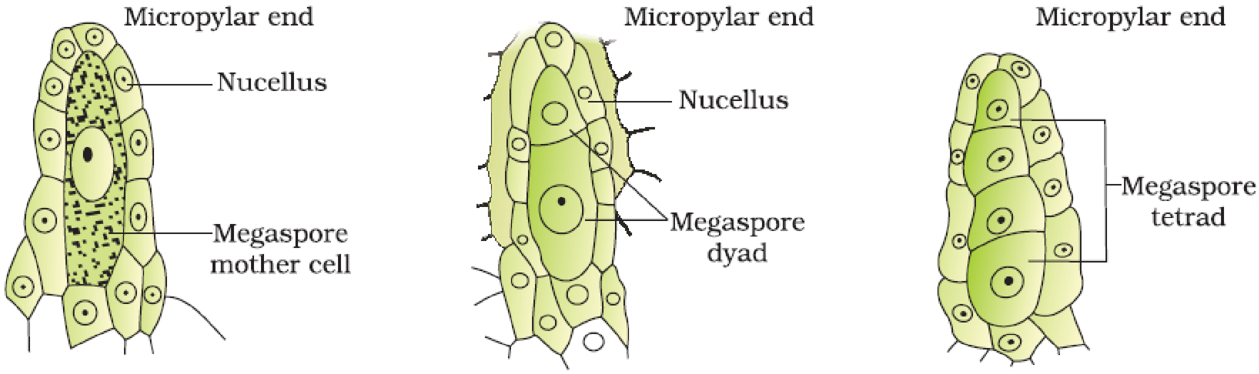


Figure: *Parts of the ovule showing a large megaspore mother cell, a dyad and a tetrad of megaspore.*

Female Gametophyte

- In most flowering plants, only one megaspore is functional while the other three degenerate.
- The single functional megaspore develops into the female gametophyte. This kind of development is called **monosporic development.**
- The nucleus of the functional megaspore divides mitotically to form 2 nuclei, which move towards the opposite ends, forming a 2-nucleate embryo sac. Two more mitotic divisions ensue, leading to the formation of 4-nucleate and 8-nucleate embryo sacs.
- After the 8-nucleate stage, the cell walls are laid down and the typical female gametophyte (embryo sac) gets organised.
- Six of the 8-nuclei get surrounded by the cell wall and the remaining two, called **polar nuclei**, are situated below the egg apparatus in the large **central cell**.
- Three of the six cells are placed at the micropylar end and constitute the **egg apparatus** (2 **synergids** + 1 **egg cell**).
- The synergids have special thickenings at the micropylar end. These are together called the **filiform apparatus**. It helps in leading the pollen tubes into the synergids.
- Three cells are at the chalazal end, and are called **antipodal cells**.
- A typical angiosperm female gametophyte is 7-celled and 8-nucleated at maturity.

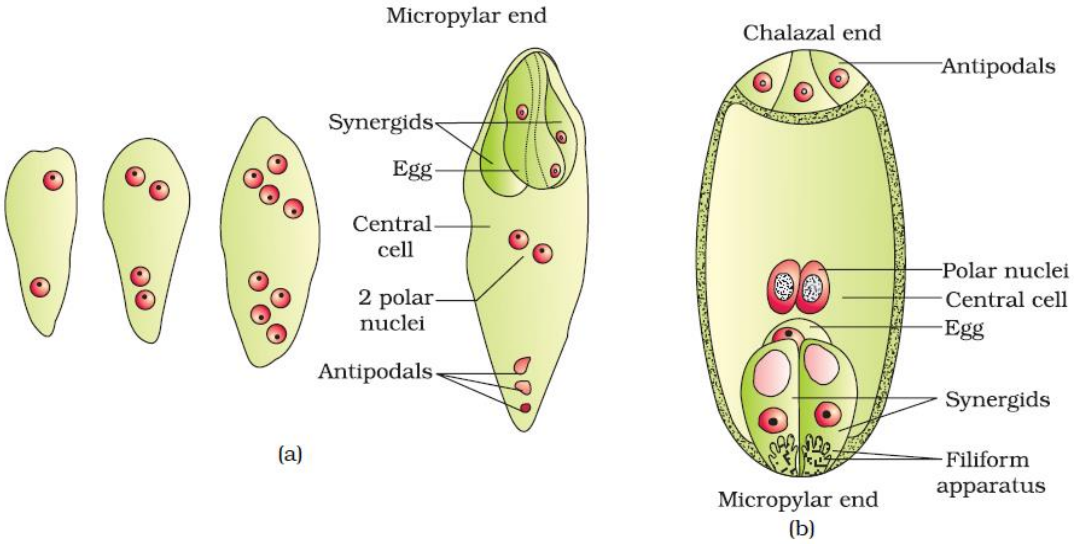
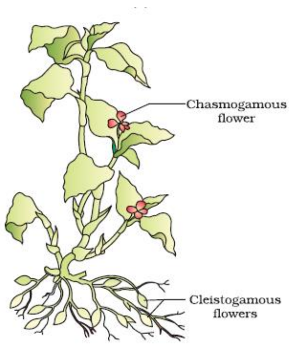


***Figure: (a)** 2, 4 and 8-nucleate stage of embryo sac and a mature embryo sac; **(b)** A diagrammatic representation of the mature embryo sac.*

Pollination



- It is the process of transfer of pollen grains from the anther to the stigma.
- Depending on the source of pollen, pollination can be divided as follows:
  - **a) Autogamy** – It is the transfer of pollen grains from the anther to the stigma of the same flower. Autogamy requires the anther and the stigma to lie close. It also requires synchrony in the pollen release and stigma receptivity. Plants like *Viola (common pansy)*, *Oxalis and Commelina* etc., produce two kinds of flowers—**chasmogamous flowers** (with exposed anther and stigma) and **cleistogamous flowers** (which do not open at all and only autogamy occurs).

Note –
- **Advantage of Cleistogamous flowers –** it produces assure seed set even in the absence of pollinators.
- **Disadvantage of Cleostogamous flowers –** it can undergo inbreeding depression and donot produce genetic variations.

  - **b) Geitonogamy** – It is the transfer of pollens from the anther of one flower to the stigma of another flower in the same plant. Genetically, it is similar to autogamy, but it requires pollinating agents.

  - **c) Xenogamy** – It is the transfer of pollen grains from the anther to the stigma of a different plant. Pollination causes genetically different types of pollens to be brought to a plant.

Agents of Pollination

- Plants use air, water (abiotic agents) and animals (biotic agents) for pollination.

- **Pollination by wind**
  - It is the most common form of abiotic pollination.
  - Plants possess well-exposed stamens and large, feathery stigma.
  - Pollens should be light and non-sticky to be carried easily by winds.
  - Wind-pollinated flowers are colourless, odourless and often have single ovule in the ovary and numerous flowers packed in an inflorescence (Ex- Corn cob – the tassels are nothing but the style and stigma which wave in the wing to trap pollen grains.)
  - It is common in grass.

***Figure:** A wind-pollinated compact inflorescence and well exposed stamens*

- **Pollination by water**
  - It is rare in flowering plants, except for some aquatic plants like fresh water plants such as *Vallisneria* and *Hydrilla* and several marine sea-grasses such as *Zostera.*
  - Flowers are not very colorful and do not produce nectar and fragnance.
  - In *Vallisneria,* the female flowers reach the surface of water by the long stalk and the male flowers or pollen grains are released on to the surface of water. They are carried passively by water currents

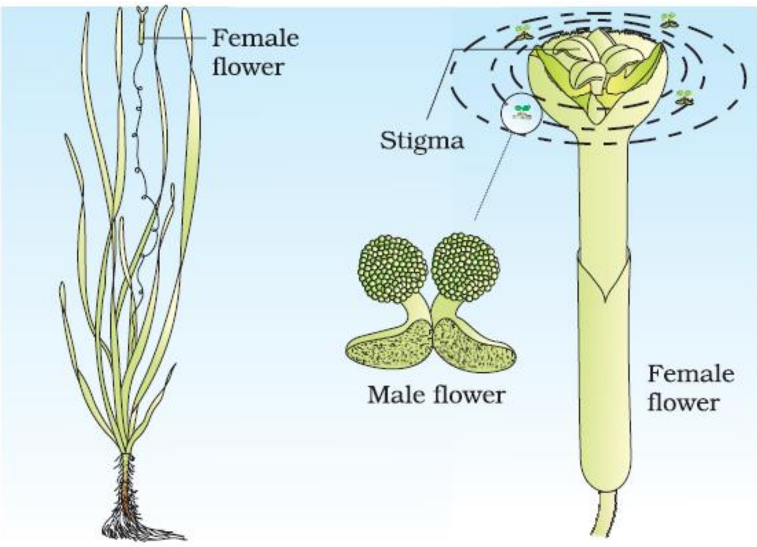


Figure: *Pollination by water in Vallisneria*

  - In *Zostera* (sea grass), female flowers remain submerged in water and the pollen grains are released inside the water. The pollen grains are long and ribbon-like
  - Most of the water pollinated species, pollen grains are protected from wetting by mucilaginous covering.

Note –
- Not all aquatic plants use water for pollination.
- In a majority of aquatic plants such as water hyacinth and water lily, the flowers emerge above the level of water and are pollinated by insect or wind as in most of the land plant.
- Both wind and water pollinated flowers are not very colorful because, the medium for pollination is not animals

- **Pollination by animals**
  - Majority of flowering plants use butterflies, bees, wasps etc., for pollination.
  - Most of the insect-pollinated flowers are large, colourful, fragrant, and contain nectar to attract the animal pollinators. These are called floral rewards (Nectar and pollen).
  - The flowers pollinated by flies and beetles secrete foul odours to attract these animals.
  - The pollen grains are sticky and get stuck to the body of the pollinator.
  - Floral reward can also be in the form of providing safe places to lay eggs (example: the tallest flower – 6 feet - *Amorphophallus*)
  - A symbiotic relationship exists between the plant, *Yucca* and its pollinator moth. The moth is dependent on the plant since the moth deposits its eggs in the locule of the ovary of the plant, and in return, the plant is pollinated by the moth.

Figure: *Insect pollination*

NOTE –
- Bees, beetles, butterflies, flies, wasps, ants, moth, birds (sunbird and humming bird) & bats are the common pollinating agents.

- Even large animals such as some primates (lemur), arboreal (tree-dwelling) rodents, or even reptiles (gecko-lizard and garden lizard) have also been reported as pollinators in some species.
- Many insects may consume pollen or the nectar without bringing about pollination. Such floral visitors are referred to as **pollen/nectar robbers.**

Out Breeding Devices

- Continued or repeated self-pollination leads to inbreeding depression.
- Plants have developed methods to prevent self pollination. Autogamy is prevented by following ways:
  - Pollen release and stigma receptivity not coordinated
  - Different positioning of the anther and the stigma
  - Self- incompatibility – is a genetic mechanism and prevents self-pollen (from the same flower or other flowers of the same plant) from fertilizing the ovules by inhibiting pollen germination or pollen tube growth in the pistil
  - Production of unisexual flowers

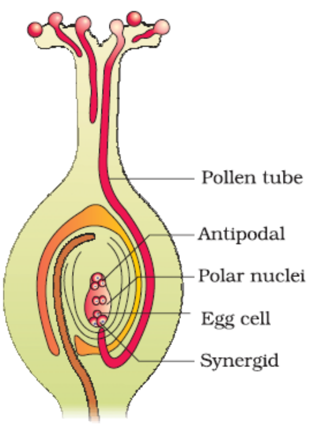
NOTE – If both male female flowers are present on the same plant such as castor and maize (monoecious), it prevents autogamy but not geitonogamy.

- Ways to prevent both autogamy and geitonogamy:
  - Presence of male and female flowers on different plants, such that each plant is either male or female (dioecy).
  - Example- Papaya.

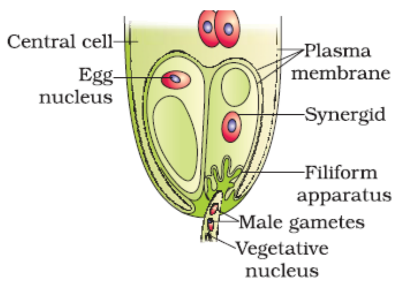
Pollen–Pistil Interactions

- Pollination does not always ensure the transfer of compatible pollens (right type).
- Hence, the pistil has the ability to recognise the right type of pollen to promote post- pollination events.
- If the pollen is of the wrong type, the pistil prevents pollen germination.
- This interaction is mediated by chemical components of the pollen and the pistil.
- Pollen–pistil interaction is a dynamic process involving pollen recognition, followed by promotion or inhibition of the pollen.
- The pollen tube reaches the ovary and enters the ovule through the micropyle. Then, through the filiform apparatus, it reaches synergids. In this way, the pollen tube grows.

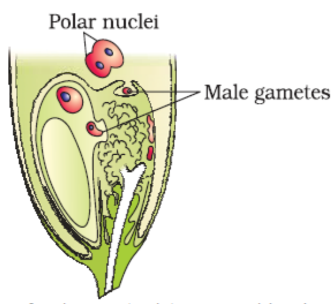
  All the events from pollen deposition on the stigma until pollen tubes enter the ovule are together referred to as pollen-pistil interaction.



Longitudinal section of a flower showing growth of pollen tube



Enlarged view of an egg apparatus showing entry of pollen tube into a synergid



Discharge of male gametes into a synergid and the movements of the sperms, one into the egg and the other into the central cell

NOTE – in some plants pollen grains are shed at two-celled condition (a vegetative cell and a generative cell). In such plants, the generative cell divides and forms the two male gametes during the growth of pollen tubes in the stigma. In plants which shed pollen in the three-celled condition, pollen tube carry the two male gametes from the beginning.

Artificial Hybridisation

- It is a method to improve crop yield.
- In this method, it is essential to ensure that the right kinds of pollen grains are used, and the stigma is protected from unwanted pollen grains. It is achieved by:
  - **Emasculation –** the anther is removed from the bud if the female parent bears bisexual flowers.
  - **Bagging –** the emasculated flower is covered by a bag (butter paper) so as not to allow contamination of the stigma by unwanted pollen grains.
- When the stigma of the bagged flower becomes receptive, the collected pollen grains are dusted onto the stigma, and then the flower is rebagged.
- If the female parent is unisexual, emasculation is not necessary. In this case, the female bud is directly bagged, and when the stigma turns receptive, suitable pollen grains are dusted onto it so as to allow germination.

Double Fertilisation

- When the pollen grains fall on the stigma, the pollen tube enters one of the synergids and releases two male gametes.
- One of the male gametes moves towards the egg cell and fuses with it to complete the **syngamy** to form the **zygote**.
- The other male gamete fuses with the two polar nuclei and forms triploid **primary endosperm nucleus (PEN)**. This is termed as **triple fusion**.
- Since two kinds of fusion—syngamy and triple fusion—take place, the process is known as double fertilisation, and is characteristic of flowering plants.
- After triple fusion, the central cell becomes the primary endosperm cell (PEC).
- The primary endosperm nucleus gives rise to the endosperm, while the zygote develops into the embryo.
- Synergids and antipodal cell degenerates.

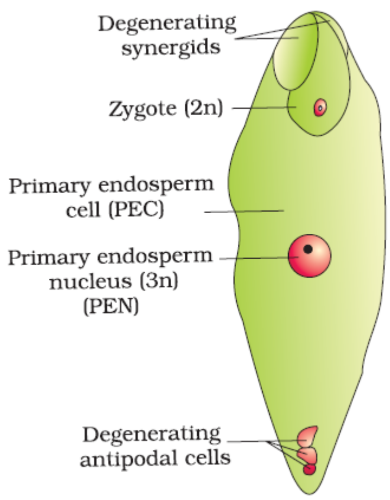


Figure: *Fertilised embryo sac showing zygote and Primary Endosperm Nucleus (PEN)*

Post-fertilisation events

It includes development of endosperm and embryo, and maturation of ovules into seeds and ovaries into fruits.

Formation of Endosperm

- The endosperm develops before the embryo because the cells of the endosperm provide nutrition to the developing embryo.
- The primary endosperm nucleus repeatedly divides to give rise to free nuclei. This stage of development is called **free nuclear endosperm** (Ex- coconut water).
- Cell wall formation occurs next, resulting in a **cellular endosperm** (Ex- White kernel surrounding coconut water).
- The endosperm may be either fully consumed by the growing embryo (as in pea, groundnut, and beans) or retained in the mature seed (as in coconut and castor).

Development of Embryo (Embryogeny or Embryogenesis)

- The embryo develops at the micropylar end of the embryo sac where the zygote is situated.
- Most zygotes divide only after the formation of endosperm. This is an adaptation to provide assured nutrition to the developing embryo.
- The zygote gives rise first to the pro-embryo, and then to the globular, heart-shaped, mature embryo.

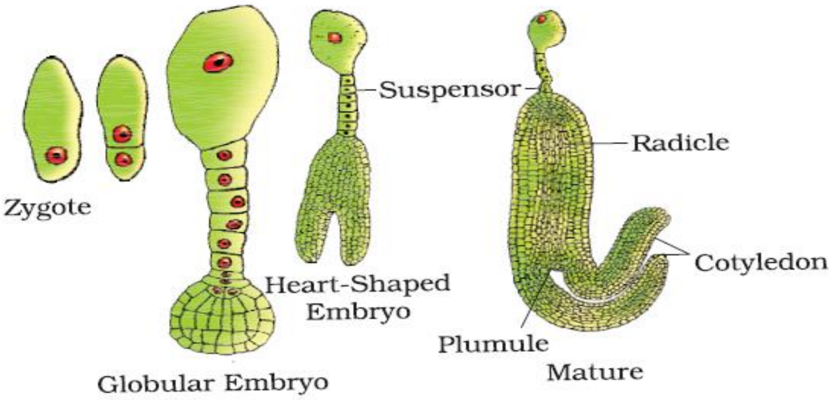


Figure: *stages in embryo development in a dicot*

- A typical **dicot embryo** consists of an embryonal axis and two cotyledons.
- The portion of the embryonal axis above the level of cotyledons is called epicotyl. It contains the plumule (shoot tip). The portion below the axis is called hypocotyl. It contains the radicle (root tip). The root tip is covered by the root cap.

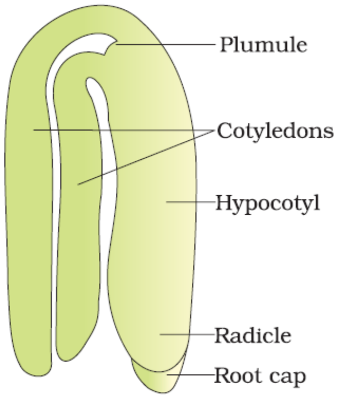


Figure: *A typical dicot embryo*

- In a **monocot embryo**, there is only one cotyledon. In grass, it is known as the scutellum, and is situated at one side of the embryonal axis. At its lower end, the embryonal axis has the radicle and the root cap enclosed in the coleorrhiza.
- The epicotyl lies above the level of the scutellum, and has the shoot apex and leaf primordia enclosed in hollow structures called coleoptiles.

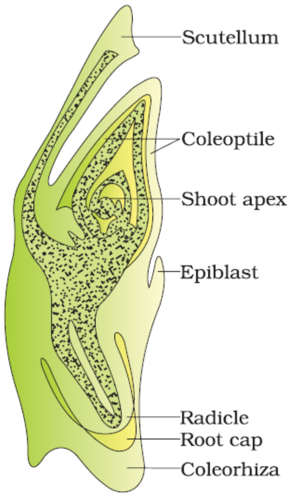


Figure: *L.S of an embryo of grass*

Development of Seeds

- It is the last stage of sexual reproduction in angiosperms.
- Seeds are the fertilised ovules that are developed inside a fruit.
- A seed consists of:
  - Seed coat
  - Cotyledons
  - Embryonal axis

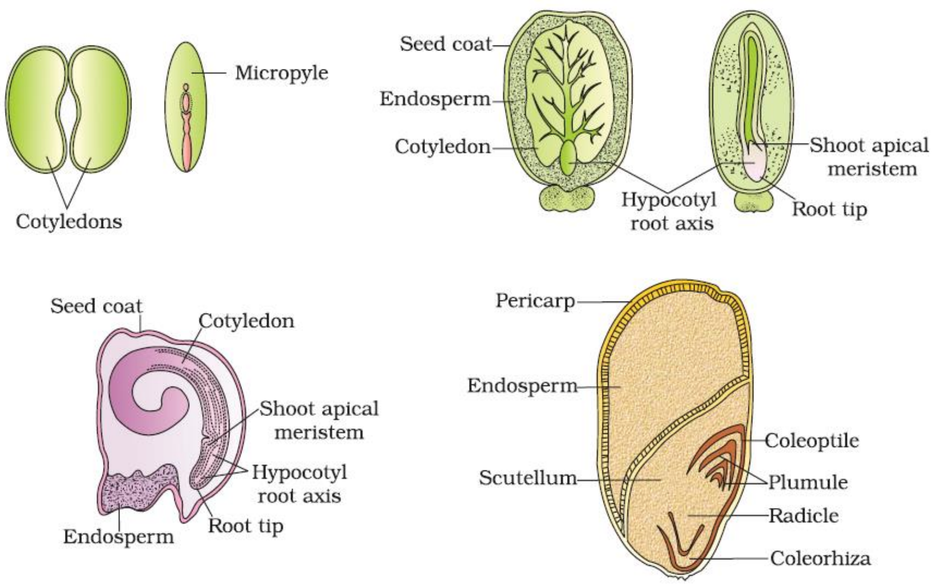


Figure: *Structure of some seeds*

- Seeds may be **albuminous** (endosperm present; as in wheat and maize) or **non-albuminous** (endosperm absent; since it is consumed by the growing embryo; as in pea, groundnut & beans).
- Some seeds such as black pepper and wheat have remnants of nucellus known as **perisperm**.
- The integuments of ovules harden to form the seed coat, and the micropyle facilitates the entry of oxygen and water into the seed.
- As it loses moisture, the seed may enter dormancy, or if favourable conditions exist, it germinates.

Advantages of seeds to angiosperms

- Seed formation in angiosperms make reproductive processes such as pollination and fertilization independent of water.
- Seeds have better strategies for dispersal to new habitats and help the species to colonise in other areas.
- Young seedlings are nourished by reserved food until they are capable of photosynthesis on their own.
- The hard seed coat provides protection to the young embryo.
- Being products of sexual reproduction, they generate new genetic combinations leading to variations.

NOTE –

- Seed is the basis of our agriculture.
- Dehydration and dormancy of mature seeds are crucial for storage of seeds which can be used as food throughout the year and also to raise crop in the next season.
- Viability of sees varies greatly between few months to hundreds of years.
- **The oldest is** that of a *lupine, **Lupinus arcticus*** excavated from **Arctic Tundra**, the seed germinated and flowered after an estimated record of **10,000 years of dormancy.**
- **A recent record of 2000 years old viable seed** is of the date palm, ***Phoenix dactylifera*** discovered during the archeological excavation at **king Herod's palace** near the Dead Sea
- *Ficus, Orchid* fruits and some parasitic species such as *Orobanche* and *Striga* contains thousands of tiny seeds.

Development of Fruits

- The ovary of a flower develops into a fruit.
- The walls of the ovary transform into the walls of the fruit (pericarp).
- Fruits may be fleshy, as in mango and orange, or can be dry, as in groundnut and mustard.
- In some plants, floral parts other than the ovary take part in fruit formation, as in apple and srawberry. In these, the thalamus contributes to fruit formation. Such fruits are called **false fruits**.

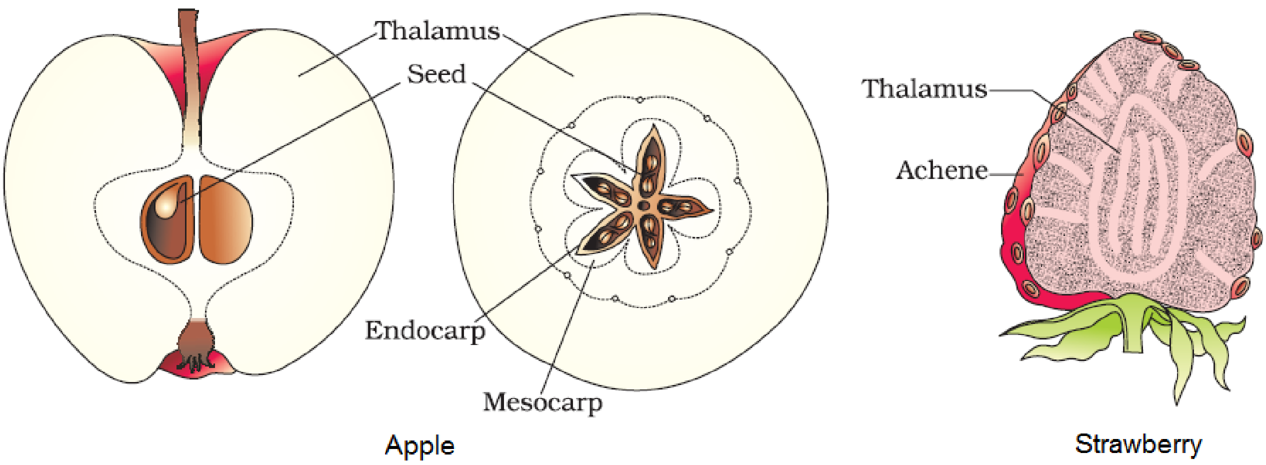


Figure: *False fruits of apple and strawberry*

- Fruits that develop from the ovary are called **true fruits**.
- Some fruits develop without fertilisation, and are known as **parthenocarpic fruits** (example: banana).

NOTE – Parthenocarpy can be induced through the application of growth hormones and such fruits are seedless

Apomixis and Polyembryony

- Some species of *Asteraceae* and grasses produce seeds without fertilisation. This process of seed formation is known as **apomixis.**
- Apomixis is a form of asexual reproduction mimicking sexual reproduction.
- In some species, apomixis occurs as the diploid egg cell is formed without meiosis, and develops into embryo without fertilisation.
- Hence, the **genetic nature** of apomictic embryos is the same as that of the parents and hence they can be called as **clone**
- In some varieties of citrus and mango, the nucellus cells divide and protrude into the embryo sac to develop into embryos. In such cases, each ovule may contain several embryos and this condition is called **polyembryony.**
- Apomixis is important for producing hybrid varieties of fruits and vegetables, and also for increasing crop yield multifold.

NOTE –

- ➢ One of the problems of hybrids is that hybrids seeds have to be produced every year. If the seeds collected from hybrids are sown, the plants in the progeny will segregate and do not maintain hybrid characters
- ➢ Production of hybrid seeds is costly and hence the cost of hybrid seeds become too expensive for the farmers.
- ➢ If these hybrids are made into apomicts, there is no segregation of characters in the hybrid progeny. Then the framers can keep on using the hybrid seeds to raise new crop year after year and he does not have to buy hybrid seeds every year.
- ➢ Because of the importance of apomixis in hybrid seed industry, active research is going on in many laboratories and to transfer apomictic genes into hybrid varieties.

SUMMARY

Flowers are the seat of sexual reproduction in angiosperms. In the flower, androecium consisting of stamens represents the male reproductive organs and gynoecium consisting of pistils represents the female reproductive organs.

A typical anther is bilobed, dithecous and tetrasporangiate. Pollen rains develop inside the microsporangia. Four wall layers, the epidermis, endothecium, middle layers and the tapetum surround the microsporangium. Cells of the sporogenous tissue lying in the centre of the microsporangium, undergo meiosis (microsporogenesis) to form tetrads of microspores. Individual microspores mature into pollen grains.

Pollen grains represents the male gametophytic generation. The pollen grains have a two-layered wall, the outer exine and inner intine. The exine is made up of sporopollenin and has germ pores. Pollen grains may have two-celled (a vegetative cell and generative cell) or three cells (a vegetative cell and two male gametes) at the time of shedding.

The pistil has three parts – the stigma, style and the ovary. Ovules are present in the ovary. The ovules have a stalk called funicle, protective integument(s), and an opening called micropyle. The central tissue is the nucellus in which the archesporium differentiates. A cell of the archesporium, the megaspore mother cell divides meiotically and one of the megaspores forms the embryo sac (the female gametophyte). The mature embryo sac is 7-celled and 8-nucleate. At the micropylar end is chalazal end are three antipodals. At the centre is a large central cell with two polar nuclei.

Pollination is the mechanism to transfer pollen grains from the anther to the stigma. Pollinating agents are either abiotic (wind and water) or biotic (animals).

Pollen-pistil interaction involves all events from the landing of pollen grains on the stigma until the pollen tube enters the embryo sac (when the pollen is compatible) or pollen inhibition (when the pollen is incompatible). Following compatible pollination, pollen grain germinates on the stigma and the resulting pollen tube grow through the style, enter the ovules and finally discharges two male gametes in one of the synergids. Angiosperms exhibit double fertilisation because two fusion events occur in each embryo sac, namely syngamy and triple fusion. The products of these fusions are the diploid zygote and the triploid primary endosperm nucleus (in the primary endosperm cell). Zygote develops into the embryo and the primary endosperm cell forms the endosperm tissue. Formation of endosperm always precedes development of the embryo.

The developing embryo passes through different stages such as the proembryo, globular and heart-shaped stages before maturation. Mature dicotyledonous embryo has two cotyledons and an embryonal axis with epicotyl and hypocotyl. Embryos of monocotyledons have a single cotyledon. After fertilisation, ovary develops into fruit and ovules develop into seeds.

A phenomenon called apomixis is found in some angiosperms, particularly in grasses. It results in the formation of seeds without fertilisation. Apomicts have several advantages in horticulture and agriculture.

Some angiosperms produce more than one embryo in their seed. This phenomenon is called polyembryony.